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Riemann Hypothesis for Projective Curves

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1 Introduction

The study of projective curves over finite fields and the Weil conjectures represents a central and fascinating topic at the intersection of algebraic geometry and number theory. A fundamental problem in this area is to understand the arithmetic properties of curves over finite fields.

Let C be a projective smooth irreducible curve over $mathbbF_q$, to systematically encode the number of points $\nu_m(C)$ of a curve over the extension fields $\mathbb{F}_{q^m}$, we introduce the zeta function of the curve, defined as the formal power series:

$$Z_C(T) = \exp\left(\sum_{m\geq 1} \nu_m(C)\frac{T^m}{m}\right) \in \mathbb{Q}[[T]]$$

In 1948, André Weil formulated a series of profound conjectures regarding the properties of this function. Among these stands the analogue of the Riemann Hypothesis for curves. As stated in Proposition 3.2, the zeta function of a smooth projective curve of genus g admits the rational form:

$$Z_C(T) = \frac{\prod_{i=1}^{2g}(1-\omega_i T)}{(1-T)(1-qT)}$$

where the inverse roots ω_i are algebraic integers. The Riemann Hypothesis specifically asserts that these roots all have an absolute value of exactly $\sqrt{q}$.

The original proof provided by Weil required advanced tools from algebraic geometry, namely the development of intersection theory on surfaces (specifically, on the product $C \times C$). In this paper, however, we will follow a different and more elementary approach.

We will present the proof published by Enrico Bombieri in 1974 [1]. This method, which refines ideas originally introduced by S. A. Stepanov, allows us to prove the Riemann Hypothesis for curves while entirely avoiding the geometry of surfaces. Instead, Bombieri's approach relies exclusively on the direct study of the function field of the curve, the Riemann-Roch theorem, and the analysis of the fixed points of the Frobenius endomorphism.

2 Basic Algebraic Geometry: Varieties and Curves

In order to understand the arithmetic properties of curves over finite fields, we must first establish the foundational geometric language. Let K be a perfect field and let $\bar{K}$ be its algebraic closure.

2.1 Projective Varieties

A point in $\mathbb{P}^n$ is written in homogeneous coordinates as $[x_0 : x_1 : \cdots : x_n]$.

Definition 2.1. Let $\bar{K}[X_0, \ldots, X_n]$ be the polynomial ring in $n+1$ variables. A *projective algebraic set* is the common zero locus in $\mathbb{P}^n$ of a collection of homogeneous polynomials. A *projective variety* V is a projective algebraic set that is irreducible (i.e., its homogeneous ideal $I(V)$ is a prime ideal in $\bar{K}[X_0, \ldots, X_n]$).

Definition 2.2. A projective variety V is said to be *defined over* K if its homogeneous ideal $I(V)$ can be generated by homogeneous polynomials with coefficients in K. In this case, we denote the set of K-rational points by $V(K)$.

2.2 Dimension and Non-singularity

The function field of a variety V, denoted by $\bar{K}(V)$, is the field of fractions of its coordinate ring.

Definition 2.3. The *dimension* of a projective variety V, denoted $\dim(V)$, is the transcendence degree of its function field $\bar{K}(V)$ over $\bar{K}$.

Definition 2.4. Let $V \subset \mathbb{P}^n$ be a variety of dimension d, and let $I(V)$ be generated by homogeneous polynomials $f_1, \ldots, f_m$. We say that V is *non-singular* (or smooth) at a point $P \in V$ if the Jacobian matrix

$$\left(\frac{\partial f_i}{\partial X_j}(P) \right)_{1 \leq i \leq m,\ 0 \leq j \leq n}$$

has rank $n-d$. If V is non-singular at every point $P \in V$, we simply say that V is a non-singular variety.

2.3 Projective Curves and Regular Functions

Definition 2.5. A *projective curve* over K is a projective variety defined over K of dimension 1. Throughout the remainder of this text, a "curve" will strictly mean a smooth, projective, one-dimensional variety.

Definition 2.6. Let C be a irreducible curve over K and $P \in C(\overline{K})$ a point. A rational function $f \in \bar{K}(C)$ is said to be *regular* at P if it can be written as $f = g/h$, where g and h are homogeneous polynomials of the same degree, and $h(P) \neq 0$.

The set of all functions regular at P forms a local ring, denoted by $\mathcal{O}_{C,P}$, with a unique maximal ideal $\mathfrak{m}_P$ consisting of the functions that vanish at P.

One of the most powerful consequences of the non-singularity of a curve is the algebraic structure of its local rings.

Proposition 2.7. *Let C be a non-singular curve and $P \in C(\overline{K})$. Then the local ring $\mathcal{O}_{C,P}$ is a Discrete Valuation Ring (DVR).*

Because $\mathcal{O}_{C,P}$ is a DVR, its maximal ideal $\mathfrak{m}_P$ is principal, meaning it is generated by a single element (not unique) $t_P \in \mathfrak{m}_P$, called a *uniformizer* at P.

Definition 2.8. Let $f \in \bar{K}(C)^\times$. Since $\mathcal{O}_{C,P}$ is a DVR with uniformizer t_P, we can uniquely write $f = u \cdot t_P^d$ for some integer $d \in \mathbb{Z}$ and some unit $u \in \mathcal{O}_{C,P}^\times$ (meaning $u(P) \neq 0$). The integer d is called the *order* of f at P, denoted by $\mathrm{ord}_P(f)$.

The function $\mathrm{ord}_P : \bar{K}(C)^\times \to \mathbb{Z}$ is a discrete valuation. It extends to $\bar{K}(C)$ by defining $\mathrm{ord}_P(0) = \infty$. If $\mathrm{ord}_P(f) > 0$, we say f has a zero at P. If $\mathrm{ord}_P(f) < 0$, we say f has a pole at P.

2.4 Divisors, Riemann-Roch, and Category Equivalence

To study the Bombieri's proof and provide the necessary tools for the proof of the Weil conjectures, we introduce the theory of divisors. This language allows us to connect the geometric points of a curve with the algebraic properties of its function field.

Definition 2.9. Let C be a smooth projective curve over K. The *divisor group* of C, denoted by $\mathrm{Div}(C)$, is the free abelian group generated by the points of C over K. A *divisor* $D \in \mathrm{Div}(C)$ is a formal finite linear combination:

$$D = \sum_{P \in C} n_P(P)$$

where $n_P \in \mathbb{Z}$ and $n_P = 0$ for all but finitely many points $P \in C$. The *degree* of D is defined as the sum of its coefficients:

$$\deg(D) = \sum_{P \in C} n_P$$

Definition 2.10. Let $f \in \bar{K}(C)^\times$ be a non-zero rational function on C. The *divisor of* f, denoted by $\mathrm{div}(f)$, is defined as:

$$\mathrm{div}(f) = \sum_{P \in C} \mathrm{ord}_P(f)(P)$$

A divisor D is called *principal* if $D = \mathrm{div}(f)$ for some $f \in \bar{K}(C)^\times$.

Since any non-zero rational function on a smooth projective curve has a total number of zeros equal to its total number of poles (counted with multiplicity), the degree of a principal divisor is always zero.

Definition 2.11. Let $D \in \mathrm{Div}(C)$ be a divisor. The *Riemann-Roch space* associated to D, denoted by $\mathcal{L}(D)$, is the $\bar{K}$-vector space defined by:

$$\mathcal{L}(D) = \{f \in \bar{K}(C)^\times \mid \mathrm{div}(f) + D \geq 0\} \cup \{0\}.$$

We denote the dimension of this space over $\bar{K}$ by $l(D) = \dim_{\bar{K}} \mathcal{L}(D)$.

The space $\mathcal{L}(D)$ consists of functions whose order of poles are bounded by the positive coefficients of D, and whose order of zeros are forced by the negative coefficients of D.

Theorem 2.12 (Riemann-Roch [2, Theorem II.5.4])**.** *Let C be a smooth projective curve of genus g, and let $K_C \in \mathrm{Div}(C)$ be a canonical divisor (the divisor associated to a non-zero differential form on C). Then for any divisor $D \in \mathrm{Div}(C)$, $l(D)$ is finite and satisfies the identity:*

$$l(D) - l(K_C - D) = \deg(D) - g + 1$$

We conclude this foundational section by stating the fundamental theorem.

Theorem 2.13. *[2, Theorem II.2.4]*
There is a contravariant equivalence of categories between:

1. *The category of smooth, projective, irreducible algebraic curves defined over K, where the morphisms are non-constant rational maps defined over K.*
2. *The category of function fields F over K of transcendence degree 1 (such that $F \cap \bar{K} = K$), where the morphisms are K-algebra homomorphisms.*

In particular, every such curve C corresponds uniquely up to K-isomorphism to its function field $K(C)$, and every K-homomorphism of function fields $\phi^* : K(C_2) \to K(C_1)$ arises from a unique non-constant morphism of curves $\phi : C_1 \to C_2$ defined over K.

3 The Zeta Function of a Curve

For the remainder of this work, we restrict our attention to the case where the base field k is a finite field of characteristic p, namely $k = \mathbb{F}_q$ with $q = p^\alpha$ ($\alpha \in \mathbb{N} \setminus \{0\}$).
Let C be a projective curve over $\mathbb{F}_q$. For any integer $m \geq 1$, we denote the number of rational points of C over the extension field $\mathbb{F}_{q^m}$ by $\nu_m(C)$.

Definition 3.1. We set

$$Z_C(T) = \exp\left(\sum_{m\geq 1} \nu_m(C)\frac{T^m}{m}\right) \in \mathbb{Q}[[T]]$$

and call it *zeta function* of C.

Rather than developing the full underlying theory, we will assume certain standard properties of this function, documented in Szamuely's notes [3], so that we may focus directly on the proof of the Riemann Hypothesis for curves.

Although we will not develop the full underlying theory, it is worth noting the mathematical genesis of this rational form. As pointed out by Bombieri [1], both the rationality of the zeta function and its functional equation are direct consequences of the Riemann-Roch theorem on C. By expanding its definition, $Z_C(T)$ can be rewritten as a generating series counting the number of effective divisors of a given degree d. For degrees $d > 2g - 2$, the Riemann-Roch theorem guarantees that the dimension of the associated Riemann-Roch spaces is exactly $d - g + 1$. This allows the infinite tail of the formal power series to be summed explicitly as a geometric series, producing the rational denominator $(1-T)(1-qT)$. The polynomial numerator of degree $2g$, on the other hand, is completely determined by the finite number of effective divisors of degree up to $2g-2$. The functional equation (Proposition 3.3) follows similarly from the duality underlying the Riemann-Roch theorem. For a detailed algebraic derivation of these properties, we refer to modern expositions such as Rosen [5].

Proposition 3.2. *If C is a smooth projective curve over $\mathbb{F}_q$ of genus g, then*

$$Z_C(T) = \frac{\prod_{i=1}^{2g}(1 - \omega_i T)}{(1 - T)(1 - qT)}$$

where the inverse roots ω_i are algebraic integers.

Proposition 3.3. *The roots of the numerator can be paired such that they satisfy the following functional equation:*

$$\omega_i \cdot \omega_{g+i} = q \quad \textit{for } i = 1, \ldots, g.$$

Relying on the rational form of the zeta function given in Proposition 3.2, one can easily give a closed formula for $\nu_r(C)$.

Proposition 3.4. *For any integer $r \geq 1$, the number of rational points over $\mathbb{F}_{q^r}$ is given by:*

$$\nu_r(C) = q^r - \sum_{i=1}^{2g} \omega_i^r + 1$$

Proof. We know that

$$\exp\left(\sum_{m\geq 1} \nu_m(C)\frac{T^m}{m}\right) = \frac{\prod_{i=1}^{2g}(1-\omega_i T)}{(1-T)(1-qT)}.$$

Taking the formal *log* on both sides of the equation we deduce that:

$$\sum_{r\geq 1} \nu_r \frac{T^r}{r} = \sum_{i=1}^{2g} \log(1-\omega_i T) - \log(1-T) - \log(1-qT).$$

Using Taylor expansion, the result follows.

□

Geometrically, the point count $\nu_r(C)$ can be interpreted via the Frobenius endomorphism. Let C be defined over $\mathbb{F}_q$; the q-th power Frobenius morphism acts on C by raising its homogeneous coordinates to the q-th power:

$$\varphi : C \to C, \quad [X_0 : \cdots : X_n] \mapsto [X_0^q : \cdots : X_n^q].$$

By the fundamental properties of finite fields, the points of C that are fixed by φ^r are precisely the rational points of C over $\mathbb{F}_{q^r}$ (since the elements of $\mathbb{F}_{q^r}$ satisfy the equation $x^{q^r} = x$). Therefore, we obtain the relation:

$$\nu_r(C) = \#\{\text{fixed points of } \varphi^r\}.$$

Our objective is to obtain an upper bound for $\nu_1(C)$. We may assume that φ has a fixed point x_0, otherwise $\nu_1(C)$ is equal to zero. Now define

$$R_m = \{f \in \bar{\mathbb{F}}_q(C)^* \mid \operatorname{div}(f) + mx_0 \geq 0\} \cup \{0\}.$$

R_m is a finite dimensional vector space on $\overline{\mathbb{F}}_q$. Recognizing that R_m is precisely the Riemann-Roch space $\mathcal{L}(mx_0)$ associated to the divisor mx_0 of degree m, we easily obtain the following bounds:

i) $\dim R_{m+1} \leq \dim R_m + 1$. Relaxing the pole condition at x_0 by exactly one order allows the dimension of the vector space to increase by at most 1.

ii) $\dim R_m \leq m+1$. This follows by induction from property (i), knowing that $\dim R_0 = 1$ (since the only regular functions without poles on a projective curve are constants). For special divisors, this upper bound is also a weak consequence of Clifford's theorem.

iii) $\dim R_m \geq m+1-g$. By the Riemann-Roch theorem, we have $l(mx_0) - l(K_C - mx_0) = m-g+1$. Since the dimension $l(K_C - mx_0)$ is always non-negative, this bound is immediate.

Since $\varphi(x_0) = x_0$, we have

iv) $R_m \circ \varphi \subseteq R_{mq}$

v) $\dim R_m \circ \varphi = \dim R_m$

vi) $\mathrm{div}(f \circ \varphi) = q \cdot \varphi(\mathrm{div(f)})$ for every $f \in R_m$

We denote by $R_l^{(p^u)}$ the subspace of R_{lp^u} consisting of functions f^{p^u} with $f \in R_l$. We note that

vii) $\dim R_l^{(p^u)} = \dim R_l$.

Lemma 3.5. *If $lp^u < q$, the natural homomorphism*

$$R_l^{(p^u)} \otimes_{\overline{\mathbb{F}}_q} (R_m \circ \varphi) \longrightarrow R_l^{(p^u)} \cdot (R_m \circ \varphi)$$

is an isomorphism.

Proof. Let $\mathrm{ord}(f)$ denote the order of a function $f \in R_m$ at the point x_0. By definition, $\mathrm{ord}(f) \geq -m$ for every $f \in R_m$. Let $r = \dim(R_m)$. By property (i), there exists a basis $s_1, \dots, s_r$ of R_m whose orders are strictly increasing:

$$\mathrm{ord}(s_i) < \mathrm{ord}(s_{i+1}).$$

To prove the lemma, it suffices to show that if $\sigma_i \in R_l$ satisfy

$$\sum_{i=1}^{r} \sigma_i^{p^u} (s_i \circ \varphi) = 0,$$

then $\sigma_i = 0$ for all $i = 1, \dots, r$.

Suppose, by contradiction, that not all σ_i are identically zero. Let $\rho \in \mathbb{N}$ be the smallest index such that $\sigma_\rho \neq 0$. Thus, $\sigma_1 = \cdots = \sigma_{\rho-1} = 0$, and the relation reduces to:

$$\sum_{i=\rho}^{r} \sigma_i^{p^u} (s_i \circ \varphi) = 0.$$

Isolating the first term, we can analyze its order. On one hand, we have:

$$\mathrm{ord}\left(\sigma_\rho^{p^u} (s_\rho \circ \varphi)\right) = p^u \, \mathrm{ord}(\sigma_\rho) + q \, \mathrm{ord}(s_\rho).$$

On the other hand, expressing this term via the remaining sum yields:

$$\begin{aligned}
\mathrm{ord}\left(\sigma_\rho^{p^u} (s_\rho \circ \varphi)\right) &= \mathrm{ord}\left(- \sum_{i=\rho+1}^{r} \sigma_i^{p^u} (s_i \circ \varphi)\right) \\
&\geq \min_{i>\rho} \mathrm{ord}\left(\sigma_i^{p^u} (s_i \circ \varphi)\right) \\
&\geq \min_{i>\rho} \left(-lp^u + q \, \mathrm{ord}(s_i)\right) \\
&= -lp^u + q \, \mathrm{ord}(s_{\rho+1}).
\end{aligned}$$

Combining these two relations, we obtain:

$$p^u \operatorname{ord}(\sigma_\rho) + q \operatorname{ord}(s_\rho) \geq -lp^u + q \operatorname{ord}(s_{\rho+1}),$$

which implies:

$$p^u \operatorname{ord}(\sigma_\rho) \geq -lp^u + q\big(\operatorname{ord}(s_{\rho+1}) - \operatorname{ord}(s_\rho)\big) > 0.$$

The strict positivity in the last step follows from the initial hypothesis $lp^u < q$ (hence $q - lp^u > 0$) and the strict monotonicity of the orders of the basis elements ($\operatorname{ord}(s_{\rho+1}) - \operatorname{ord}(s_\rho) \geq 1$).

This shows that $\operatorname{ord}(\sigma_\rho) > 0$. However, $\sigma_\rho \in R_l$, meaning it has no poles outside of x_0. A rational function with no poles everywhere and at least one zero must be identically zero. This yields $\sigma_\rho = 0$, which contradicts our initial assumption.

□

Corollary 3.6. *If $lp^u < q$ then*

$$\dim R_l^{p^u}(R_m \circ \varphi) = (\dim R_l)(\dim R_m)$$

Proof. By (v) $\dim R_m \circ \varphi = \dim R_m$ and by (vii) $\dim R_l^{p^u} = \dim R_l$. Using the Lemma we obtain

$$\dim R_m \dim R_l = \dim(R_m \circ \varphi) \dim R_l^{(p^u)} = \dim R_l^{(p^u)}(R_m \circ \varphi).$$

□

Now we can finally give an upper bound for $\nu_1(C)$.

Theorem 3.7. *Assume $q = p^\alpha$, with α even. Then if $q > (g+1)^4$ we have*

$$\nu_1 < q + (2g+1)q^{\frac{1}{2}} + 1.$$

Proof. Assuming $lp^u < q$, Lemma 3.5 ensures that the morphism

$$\delta : R_l^{(p^u)}(R_m \circ \varphi) \longrightarrow R_l^{(p^u)} R_m \subseteq R_{lp^u+m}$$

defined by the mapping $\sum \sigma_i^{p^u}(s_i \circ \varphi) \mapsto \sum \sigma_i^{p^u} s_i$ is well-defined. Furthermore, applying Corollary 3.6 and the Riemann-Roch theorem, for any $l, m \geq g$ we obtain:

$$\begin{aligned} \dim \ker(\delta) &\geq \dim R_l \dim R_m - \dim R_{lp^u+m} \\ &\geq (l+1-g)(m+1-g) - (lp^u + m + 1 - g) \end{aligned}$$

We aim to choose parameters l, m, and u such that $\dim \ker(\delta) > 0$. Let $f \in \ker(\delta) \setminus \{0\}$. For every fixed point x of φ (with the possible exception of x_0), we have $\operatorname{ord}_x(f) \geq p^u$. Indeed,

$$\begin{aligned} f(x) &= \sum \sigma_i^{p^u}(x) s_i(\varphi(x)) \\ &= \sum \sigma_i^{p^u}(x) s_i(x) \\ &= \delta f(x) = 0. \end{aligned}$$

Consequently, f has at least $p^u(\nu_1 - 1)$ zeros. Moreover, since $f \in R_{lp^u+mq}$, it has at most $lp^u + mq$ poles. Since the total number of zeros of f must equal its total number of poles, it follows that $(\nu_1 - 1)p^u \leq lp^u + mq$. Therefore,

$$\nu_1 \leq l + \frac{mq}{p^u} + 1. \tag{1}$$

Let us make the specific choice:

$$u = \frac{\alpha}{2}, \quad m = p^u + 2g \quad \text{and} \quad l = \left\lfloor \frac{g}{g+1} p^u \right\rfloor + g + 1.$$

It is straightforward to verify that this choice satisfies $l, m \geq g$, $lp^u < q$, and $\dim\ker(\delta) > 0$, thus ensuring that inequality (1) holds. Substituting these values into the inequality yields:

$$\nu_1 \leq \frac{g}{g+1} q^{\frac{1}{2}} + g + 1 + q + 2g + q^{\frac{1}{2}} + 1 = q + 2gq^{\frac{1}{2}} + \frac{g}{g+1} q^{\frac{1}{2}} + g + 2.$$

To obtain the desired upper bound, it remains only to show that:

$$\frac{g}{g+1} q^{\frac{1}{2}} + g + 2 \overset{?}{<} q^{\frac{1}{2}} + 1$$

This is equivalent to:

$$g + 1 \overset{?}{<} q^{\frac{1}{2}} \left(1 - \frac{g}{g+1}\right) \iff (g+1)^2 \overset{?}{<} q^{\frac{1}{2}}$$

This final inequality is guaranteed by our initial hypothesis. Thus, we conclude that:

$$\nu_1 \leq q + 2gq^{\frac{1}{2}} + \frac{g}{g+1} q^{\frac{1}{2}} + g + 2 < q + 2gq^{\frac{1}{2}} + q^{\frac{1}{2}} + 1.$$

□

4 Proof of the Riemann Hypothesis for Curves

With Theorem 3.7 we have given an upper bound to $\nu_1(C)$, but to prove the Riemann Hypothesis we will also need a lower bound. This lower bound will be a consequence of the following Lemma.

Lemma 4.1.

$$\nu_1(C) = q + O(q^{\frac{1}{2}}).$$

Remark on notation: Here, the O symbol is used to denote an asymptotic upper bound with respect to the variable q. Specifically, it means that there exists a positive constant c, which depends only on the geometric properties of the curve C (such as its genus) and is strictly independent of q, such that $|\nu_1(C) - q| \leq cq^{1/2}$.

Proof. Consider the function field $\overline{k}(C)$ of the curve $C/\overline{k}$. It contains a purely transcendental subfield $\overline{k}(t)$ such that $\overline{k}(C)$ is a separable extension of $\overline{k}(t)$. Hence there is a normal extension $\mathbb{F}$ of $\overline{k}(t)$ which is also normal over $\overline{k}(C)$.

By Theorem 2.13, there exists a projective curve C' such that $\mathbb{F} = \overline{k}(C')$. So we have

$$C' \to C \to \mathbb{P}^1$$

where $C' \to \mathbb{P}^1$ is Galois, with Galois group G and $C' \to C$ is also a Galois covering, corresponding to a subgroup H of G.

If x is a point of $\mathbb{P}^1$ rational over k and unramified in $C' \to \mathbb{P}^1$, and if y is a point of C' lying over x, since the action of G on the fibers is transitive, we have

$$\varphi(y) = \eta \cdot y$$

for some $\eta \in G$, called the Frobenius substitution of G at the point y.

Let $\nu_1(C', \eta)$ be the number of such points of C' with Frobenius substitution η. By the same argument as Theorem 3.7 but using the map

$$\delta_\eta : R_l^{(p^\mu)}(R_m \circ \varphi) \to R_l^{(p^\mu)}(R_m \circ \eta)$$

instead of δ, we obtain easily

$$\nu_1(C', \eta) \leq q + (2g' + 1)q^{\frac{1}{2}} + 1 \tag{2}$$

where g' is the genus of C'.

On the other hand

$$\sum_{\eta \in G} \nu_1(C', \eta) = |G|\nu_1(\mathbb{P}^1) + O(1) \tag{3}$$

where $O(1)$ is added to take care of the branch points of $C' \to \mathbb{P}^1$.
Since $\nu_1(\mathbb{P}^1) = q + 1$, using (2) and (3), we deduce that

$$\nu_1(C', \eta) = q + O(q^{\frac{1}{2}}) \tag{4}$$

for every $\eta \in G$.
Indeed, let $E_\eta = \nu_1(C', \eta) - q$ denote the error term for a given $\eta \in G$.

From Equation (2), since the genus g' is constant, it can be added in $O(q^{1/2})$, so we have the upper bound:

$$E_\eta \leq O(q^{1/2}) \quad \forall \eta \in G$$

Equation (3) states that $\sum_{\eta \in G} \nu_1(C', \eta) = |G|\nu_1(\mathbb{P}^1) + O(1)$. Substituting $\nu_1(\mathbb{P}^1) = q + 1$, we obtain:

$$\sum_{\eta \in G} (q + E_\eta) = |G|(q + 1) + O(1).$$

Expanding the sum yields $|G|q + \sum_{\eta \in G} E_\eta = |G|q + |G| + O(1)$. Since $|G|$ is a constant, it is absorbed by the $O(1)$ term:

$$\sum_{\eta \in G} E_\eta = O(1).$$

To find the lower bound for a specific element η_0, we isolate E_{η_0} from the total sum:

$$E_{\eta_0} = \sum_{\eta \in G} E_\eta - \sum_{\eta \neq \eta_0} E_\eta = O(1) - \sum_{\eta \neq \eta_0} E_\eta.$$

By the upper bound established from (2), there exists a positive constant c such that $E_\eta \leq cq^{1/2}$ for all $\eta \in G$. The sum of the remaining $|G| - 1$ terms is therefore bounded above by $(|G| - 1)cq^{1/2} = O(q^{1/2})$. Substituting this into our equation yields:

$$E_{\eta_0} \geq O(1) - O(q^{1/2}) = -O(q^{1/2}).$$

Having established that E_{η_0} is bounded from above by $cq^{1/2}$ and from below by $-Cq^{1/2}$ (for some positive constants c, C), it follows that its absolute value is bounded by a constant multiple of $q^{1/2}$, which means exactly $E_{\eta_0} = O(q^{1/2})$. Thus, we obtain (4):

$$\nu_1(C', \eta) = q + O(q^{1/2}).$$

Now, let us relate this back to the curve C. By applying the point counting argument to the Galois covering $C' \to C$ with Galois group $H \subseteq G$, we have the relation:

$$\sum_{\eta \in H} \nu_1(C', \eta) = |H|\nu_1(C) + O(1).$$

On the other hand, substituting the estimate (4) into the sum over H, we obtain:

$$\sum_{\eta \in H} \nu_1(C', \eta) = \sum_{\eta \in H} \left(q + O(q^{1/2})\right) = |H|q + O(q^{1/2}).$$

Equating these two expressions gives:

$$|H|\nu_1(C) + O(1) = |H|q + O(q^{1/2}).$$

Since the order $|H|$ of the subgroup is a constant independent of q, we can divide both sides by $|H|$ and absorb the $O(1)$ term into the $O(q^{1/2})$ term. This yields the desired thesis:

$$\nu_1(C) = q + O(q^{1/2}).$$

□

Theorem 4.2 (Riemann Hypothesis for Projective Curve)**.** *The absolute value of the inverse roots of the Zeta function of a projective curve C/k is equal to $q^{\frac{1}{2}}$.*

Proof. From Proposition 3.4, the number of rational points of the curve C over the extension field $\mathbb{F}_{q^r}$ is exactly given by:

$$\nu_r(C) = q^r + 1 - \sum_{i=1}^{2g} \omega_i^r$$

where ω_i are the inverse roots of the Zeta function.

By extending the result of Lemma 4.1 to the extension field $\mathbb{F}_{q^n}$, we have the upper bound:

$$|\nu_n(C) - (q^n + 1)| \leq c \cdot q^{n/2}$$

for some positive constant c independent of n. So we obtain:

$$\left| \sum_{i=1}^{2g} \omega_i^n \right| \leq c \cdot q^{n/2} \tag{5}$$

We proceed by contradiction to show that all roots have absolute value is less or equal then $q^{1/2}$.
Since the number of roots is finite, we can define $M = \max_{1 \leq i \leq 2g} |\omega_i|$ as their maximum absolute value. Assume, by contradiction, that $M > q^{1/2}$.

Let $k \geq 1$ be the exact number of roots that achieve this maximal modulus M. Up to reordering, we can assume these are $\omega_1, \ldots, \omega_k$. Writing them in polar form, we have $\omega_j = Me^{i\theta_j}$ for $j = 1, \ldots, k$. The remaining $2g - k$ roots have a modulus strictly bounded by some real number $M' < M$.

By Dirichlet's Theorem on Simultaneous Diophantine Approximation [6, Theorem 200], for any arbitrarily small $\epsilon > 0$, there exists a sequence of arbitrarily large integers n such that the angles $n\theta_j$ are ϵ-close to an integer multiple of 2π for all $j = 1, \ldots, k$. For these chosen exponents n, the cosine of $n\theta_j$ is bounded below by $\cos \epsilon$. Thus, the real part of the sum of the n-th powers of the maximal roots satisfies:

$$\mathrm{Re} \left(\sum_{j=1}^{k} \omega_j^n \right) = \sum_{j=1}^{k} M^n \cos(n\theta_j) \geq kM^n \cos \epsilon.$$

On the other hand, the sum of the n-th powers of the remaining strictly smaller roots is bounded by $(2g - k)(M')^n$. By the triangle inequality, the total sum is bounded from below by:

$$\left| \sum_{i=1}^{2g} \omega_i^n \right| \geq \mathrm{Re} \left(\sum_{i=1}^{2g} \omega_i^n \right) \geq kM^n \cos \epsilon - (2g - k)(M')^n.$$

If we choose ϵ small enough so that $\cos \epsilon > 1/2$, and we take n sufficiently large within our sequence, the term $(M')^n$ becomes negligible compared to M^n. This yields the lower bound:

$$\left| \sum_{i=1}^{2g} \omega_i^n \right| \geq \frac{k}{2} M^n.$$

Substituting this lower bound back into inequality (5), we obtain:

$$\frac{k}{2} M^n \leq c \cdot q^{n/2}.$$

Since we assumed $M > q^{1/2}$, the left-hand side of this inequality grows strictly faster than the right-hand side with respect to n. For a sufficiently large integer n, inequality (5) is definitively violated. This provides a contradiction. Therefore, we must have:

$$|\omega_i| \leq q^{1/2} \quad \text{for all } i = 1, \ldots, 2g.$$

Finally, Proposition 3.3 provides the functional equation for the roots:

$$\omega_i \cdot \omega_{g+i} = q$$

Taking the absolute value, we get $|\omega_i||\omega_{g+i}| = q$. Since all the roots are bounded by $q^{1/2}$, the only remaining possibility is that:

$$|\omega_i| = q^{1/2} \quad \text{for all } i = 1, \ldots, 2g$$

Concluding the proof of the theorem.

□

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